A Densely Attentive Refinement Network for Change Detection Based on Very-High-Resolution Bitemporal Remote Sensing Images

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***Abstract*— Detecting changes using bitemporal remote sensing imagery is vital to understand the dynamics of the land surface. Existing change detection models based on deep learning suffer from the problem of scale variation and pseudochange due to their insufficient multilevel aggregation and inadequate capability of feature representation, which limits the accuracy. This study proposes a densely attentive refinement network (DARNet) to improve change detection on bitemporal very-high-resolution remote sensing images. DARNet is based on the U-shape encoder–decoder architecture with the Siamese network as a feature extractor. The dense skip connection module (DSCM) is employed between the decoder and the encoder to aggregate multilevel feature maps. The hybrid attention module (HAM) is integrated to exploit contextual information and generate discriminative features. The recurrent refinement module (RRM) is exploited to progressively refine the predicted change maps during the decoding process. Experiments on testing the model performance were conducted on three benchmark datasets: the season-varying change detection (SVCD) dataset, the Sun Yat-sen University change detection (SYSU-CD) dataset, and the Learning Vision and Remote Sensing Laboratory building change detection (LEVIR-CD) dataset. The experimental results demonstrate that DARNet outperforms state-of-the-art models with kappa of 96.58%, 75.35%, and 90.69% for the SVCD, SYSU-CD, and LEVIR-CD datasets, respectively.**

***Index Terms*— Attentio, bitemporal images, hange detection, refinement, skip connection.**

I. Introduction

CHANGE detection aims at detecting and extracting the changes of objects in the same geographic area using multitemporal remote sensing data [1]. It has been widely applied in different remote sensing fields, such as land management [2], disaster assessment [3], and urban expansion investigation [4]. With the development of spaceborne and airborne sensors (e.g., Gaofen-2, GeoEye, and WorldView), large amounts of high-resolution remote sensing data become available for potential monitoring of land surface changes. Since the manual interpretation of multitemporal images is both time- and labor-consuming, researchers attempt to develop automatic methods for identifying the changed area accurately and efficiently.

A number of change detection methods have been proposed in recent years. Common methods include the pixel- and object-based methods [5], [6]. The idea for the pixel-based methods is to calculate the variation map by direct comparisons of pixels in multitemporal images and then obtain the change maps using the algorithms such as threshold segmentation and clustering. Methods such as algebra-based methods, transformation-based methods, and classification-based methods have been devised to excavate the features of pixels from the images acquired at different times. The algebra-based methods apply arithmetical operations, such as image differencing [7], [8] and image rationing [9], to derive the change maps. The transformation-based methods, such as the principal component analysis [10], the change vector analysis [11], and the tasseled cap transformation [12], transform the features into a specific feature space to highlight the changing information in images. Classification-based methods distinguished the changed pixels by comparing the classification results produced using remote sensing images acquired at different time or using a trained classifier to identify the class of changed areas in multitemporal images directly [6]. Since the performance of the classification-based methods highly depends on the effectiveness of the used classifiers, there are studies to test different machine learning-based classifiers, such as the support vector machine (SVM) [13]. The pixel-based methods largely focus on the features of the individual pixel but overlook the contextual information, making it prone to noise in the detection results, such as isolated changed pixels, holes in the changed area, or jagged boundaries [14]. These conventional pixel-based methods are mostly applied to coarse- and moderate-resolution remote sensing images and show limited accuracy in high-resolution images due to increased variability [5], [15]. As a result, the pixel-based methods could produce considerable omission and commission errors because of the pseudochange caused by external factors, such as geometric registration error, illumination difference, and scale variance [16]. To overcome these problems, scholars introduced the method of object-based image analysis to detect

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the changed area in remote sensing [17], [18] and attempted to analyze objects rather than individual pixels in the remote sensing data. The object-based methods divide multitemporal images into many homogeneous regions based on the similarity of spectral and spatial adjacency and then exploit spatial contextual information, such as texture, shape, and the spatial relationship with neighbor objects [19], [20]. With the integration of abundant features, the map of changed areas is often generated using the postclassification strategy [21]. The handcrafted features have insufficient representation ability of high-level features, which greatly hampers the accuracy of models. Moreover, the object segmentation strategy largely influences the performance of change detection, which means that the error caused by the segmentation would propagate to the predicted change maps.

With the continuous development of the deep learning models in many computer vision tasks, deep learning models have also been applied to deal with tasks in remote sensing image interpretation, such as scene classification, geographical object detection, and change detection [22]–[25]. Because deep learning models, such as the convolutional neural network (CNN), have excellent abilities on capturing and representing the informative image features, the deep learning models improve the performance of various tasks in remote sensing image interpretation largely. In addition to the CNN, other state-of-the-art (SOTA) methods are also devised and applied in remote sensing. For instance, generative adversarial networks (GANs) enable higher robustness of features and mitigation of discontinuous problems by adversarial learning [26], [27]. Graph convolutional networks (GCNs) with great ability to handle graph data can be used to model long-range spatial relations [28], [29]. Very recently, the emerging model, Transformer, gains enormous popularity due to its efficacy in analyzing sequential data using self-attention block, which has shown significant improvement in hyperspectral image classification [30]. Besides, the multimodal learning methods address the limitation of information diversity and exploit abundant characteristics from multimodal remote sensing data [31]. In the field of remote sensing change detection, CNN-based models have achieved better performance than traditional methods on remote sensing change detection because CNN is able to capture regional information and does not need any handcrafted features. Deep learning-based change detection can be further divided into the patch- and image-based models. Patch-based methods use small image patches (e.g., 3×3, 5×5, and 7×7) as input to distinguish the category of central pixel [32]. Although such methods do not require amounts of data for training, applying them is time-consuming and has very low computational efficiency due to the overlap between adjacent patches [16]. Therefore, the mainstream deep-learning-based methods are still image-based because they can conduct image-to-image prediction in an end-to-end and efficient way. Among a variety of Image-based models, fully convolutional network (FCN) [33] and its variants gain increasing popularity. Based on the way to tackle multitemporal images, the CNN-based models for change detection generally adopt early fusion (EF) and late-fusion (LF) methods. The basic idea for EF methods is to treat the concatenations or differences of multitemporal images as model inputs. Alcantarilla *et al.* [34] concatenated the bitemporal images with RGB bands as a composite image with six bands and fed the concatenated images into an FCN to obtain binary change maps. Lei *et al.* [35] used a difference map of bitemporal images preprocessed by multivariate morphological reconstruction as model inputs and then proposed an FCN with pyramid pooling to detect changes in landslide inventory. Compared with EF methods that use an individual input, LF methods use images acquired at different times as separated inputs. Discriminative features of images acquired at different times are extracted by deep CNN separately and then fused or compared to generate change maps. Zhan *et al.* [36] utilized a Siamese network that adopts a metric learning strategy, and the initial results produced by simple threshold segmentation were updated by the k-nearest neighbor approach. Daudt *et al.* [37] designed two LF models based on Siamese extension of FCN to perform change detection. The encoders of both models are extracted and fused multilevel features at each time phase, whereas the decoders discriminated the changed area. In comparison with EF models, the experimental results indicated that the LF strategy often gains better performance.

Although great improvement has been brought by deep learning methods, there still exist some shortcomings that hinder the performance to be further improved. First, as the spatial resolution of remote sensing images keeps rising, more tiny objects and spatial details can be observed. However, the existing methods cannot effectively deal with the scale variation problem, especially the locating small sample and capturing detailed characteristics of changed objects due to the loss of fine-resolution features in the prevalent architecture with successive downsampling when encoding high-level features. An effective method that can recover spatial detail and preserve high-level semantic information is required. Second, the complexity of the geographical object and environmental factors of optical imaging poses a great challenge to the development of change detection models. Due to the inadequate capability in features extraction of existing models, it is still challenging to well recognize the semantic change of interested objects and filter out irrelevant changes. It also easily resulted in a serious pseudochange problem because the extracted features lack the robustness to the geometric misregistration and spectral differences caused by illumination, noise, and seasonality of creatures between bitemporal images. Thereby, more effective techniques should be devised to extract features with more discriminative information to help distinguish the hard sample from bitemporal images. Third, most of the networks in previous works were designed to be more complex and deeper because they can theoretically improve feature learning when sufficient data are provided. However, the increasing complexity not only increases the computational cost during training and inferencing but also raises the probability of overfitting and vanishing gradients, which undermines the feature representation and, thus, undermines the performance.

To deal with the aforementioned problems, a densely attentive refinement network (DARNet) is proposed for

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change detection using very-high-resolution bitemporal remote sensing images. The main contributions of this research are given as follows.

1) We develop a novel DARNet for conducting pixel-level change detection prediction in an end-to-end manner. The network adopts a simple Siamese FCN network with shared weights to extract multilevel and multiscale feature representation in different phases for bitemporal images.
2) We devise three effective modules in DARNet for advancing the performance of change detection. The dense skip connection module DSCM) aggregates features from different levels and with different scales, which enables detection of changed objects with various scales and recovering spatial details in the decoding stage. Thereinafter, the hybrid attention module (HAM) comprising the efficient spatial–temporal attention module (ESTAM) and the channel attention module (CAM) can effectively fuse multilevel features and captures long-range relations from spatial, temporal, and channel dimensions, generating more discriminative features for change detection. Finally, we leverage the recurrent refinement module (RRM) to generate stagewise predictions where the former predictions provide prior information for latter predictions. It not only conducts coarse-to-fine refinement of change maps in an efficient way but also alleviates the problem of overfitting and vanishing gradients.
3) We carry out extensive experiments on three benchmark datasets to comprehensively evaluate the performance of the proposed DARNet on remote sensing change detection tasks from various aspects. By comparing with other SOTA models quantitatively and qualitatively, DARNet not only demonstrates a prominent superiority in terms of detection accuracy but also attains a better tradeoff between accuracy and complexity.

II. Related Works

A. Skip Connection on Change Detection Networks

Common semantic segmentation models followed the U-shape architecture that contains consecutive downsampling layers on the encoder and upsampling layers on the decoder. The intermediate features' map contains abundant information that is conducive to discriminate the class of pixels. Skip connection is one of the most effective structures that can take advantage of these rich feature maps for improving performance on pixelwise predictions.

U-Net initially introduced the skip connection into the encoder–decoder architecture on semantic segmentation tasks. Following the U-Net's design, such plain connections between the encoder and the decoder are also exploited in remote sensing change detection [37]–[39]. It helps recover the spatial details when the feature maps are upsampled in the decoding stage, whereas it is still difficult to detect multiscale change objects with well-preserved details. To overcome the drawback, various variants of U-Net were proposed in previous works [15], [40], [41]. Yu *et al.* [40] proposed NestNet based on UNet++, which used a parallel module to extract features of images and utilized absolute different operations to process bitemporal features. Peng *et al.* [15] directly combined coregistered image pairs and forwarded them to an improved UNet++ network for change detection. The used UNet++ architecture contains a series of nested connections, which helps to recover fine-grained details and preserves semantic information for change detection. These dense connections can help the model capture multiscale and multilevel features for fusion and mitigate pseudochange problems caused by scale variance. Despite the achievements, the connection strategy used in UNet++ is incapable of exploiting sufficient information from full scales within the network. There is room for scholars to enhance the change detection performance by incorporating full-scale features and further narrowing the semantic gap between the encoder and the decoder at each individual level.

B. Attention Mechanism on Change Detection Networks

The attention mechanism in computer vision originates from the instinct of human vision, which selectively pays attention to a certain portion of the whole image but suppresses irrelevant information [42]. As the context information is important for generating highly discriminative features in the task of remote sensing change detection, attention mechanism has been widely employed in change detection networks [43]–[46]. By introducing the attention mechanism, the models could exploit long-range dependencies and produce discriminative context features, thereby improving the detection accuracy and model robustness to pseudochange problems [16], [45], [47]. Chen *et al.* [47] introduced a self-attention mechanism capturing channelwise and spatialwise attention into an FCN with the Siamese network such that the model could achieve robust change detection. Wang *et al.* [46] employed an adaptive attention mechanism adapted from the convolutional block attention module (CBAM) [48] to emphasize relevant features in spatial and channel domains. These studies and some other attention-based methods mainly focus on modeling the long-range dependencies on the spatial or channel domain, and they overlook the temporal relationship among images acquired at different times, while the temporal relationship could play an important role in change detection rather than semantic segmentation. It is quite possible that the models could generate discriminative image features for change detection if the temporal information is exploited. To model temporal dependency, Chen and Shi [45] designed a novel module that exploits spatial–temporal relationship of pixels at different times and positions. Although the captured spatial–temporal dependencies make the model robust to the variance of illumination and misregistration, the computation of self-attention has high computing complexity and requires considerable amounts of GPU memory consumption. This means that a more efficient and effective attention module that can model long-range dependencies in different dimensions is needed for more robust change detection from multitemporal remote sensing images.

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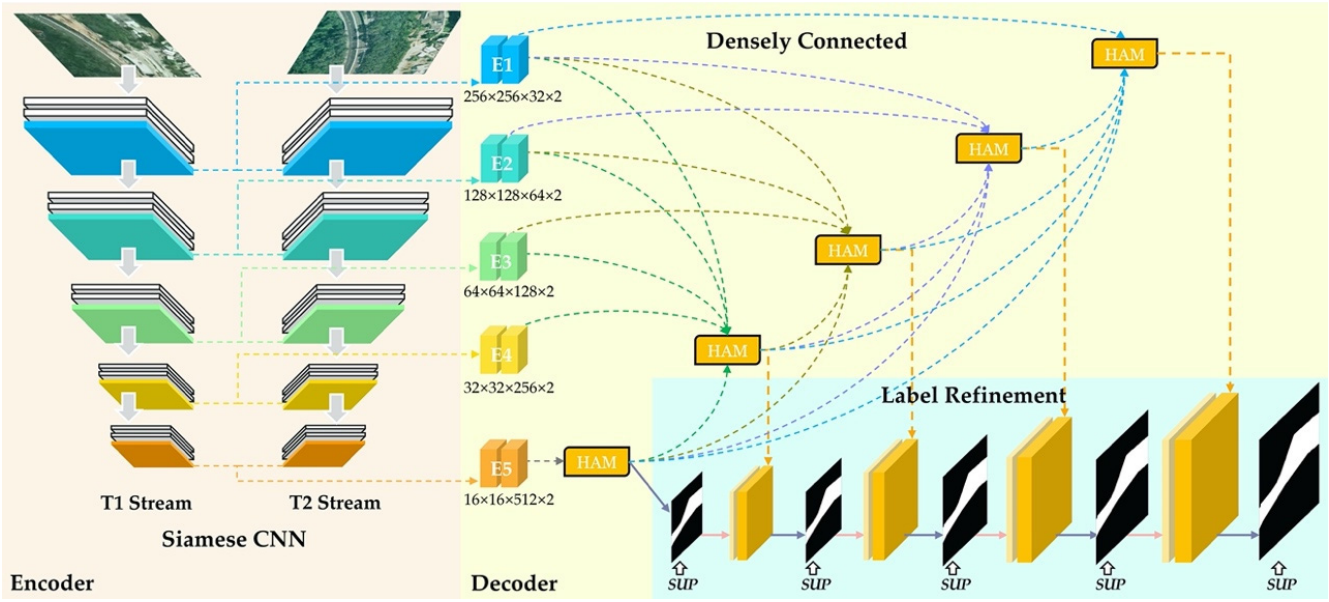

Fig. 1. Architecture of the DARNet, where E1–E5 denote the bitemporal feature maps extracted by the encoder, HAM denotes the hybrid attention module, and SUP represents deep supervision for each predicted change map.

III. Methodology

A. Overview

As shown in Fig. 1, DARNet utilizes a typical encoder–decoder architecture that is composed of three parts: the encoder, the decoder, and an extra refinement module. Given a pair of bitemporal images covering the same place, the encoder first generates multiscale features to learn discriminative features for change detection. DSCM is then utilized to connect features from decoding layers and encoding layers so that the network can fully excavate multiscale features at different levels. These multiscale features are forwarded into the HAM to make the features more discriminative by modeling nonlocal contextual information on spatial, temporal, and channel domains. An RRM with deep supervision is appended to the output layers in the decoder, thereby refining the prediction results stage by stage. Moreover, the hybrid objective function is employed to compute the loss between predictions in each deep-supervised layer and the corresponding ground truth, after which these multistage losses were aggregated to facilitate convergence during the model training.

B. Feature Extractor

In the encoder, we employed a Siamese convolution neural network to extract features with different scales and levels from image pairs. For effective extraction of representative temporal information in the same feature space, the network is composed of two identical subnetworks that have the same weights, and each subnetwork extracts deep features from the corresponding time-phase image separately. We design a simple backbone network to serve as the feature extracting subnetwork, and the backbone network consists of five convolution blocks, as shown in Fig. 1. The output channels of the five blocks are 32, 64, 128, 256, and 512, respectively. All five blocks have the same architecture that consists of two convolution modules, including a 3×3 convolution layer, a BatchNorm layer [49], and an ReLU activation layer [50]. The max-pooling layer with a stride of 2 is added between every two convolution blocks to half the size of the input. As the network gets deeper, features are projected to higher dimensions for high-level feature extraction. This encoder could generate five pairs of feature maps at different scales, which are denoted as F^1_{En}, F^2_{En}, F^3_{En}, F^4_{En}, and F^5_{En} with the sizes of 1, 1/2, 1/4, 1/8, and 1/16 of the input image. Note that the designed backbone network is relatively simpler than the other benchmark network, such as VGGNet [51] and ResNet [52]. One reason is that we aim to illustrate that DARNet can achieve the SOTA performance even with a simple feature extractor and, thus, highlight the effectiveness of the novel network components.

C. Dense Skip Connection Module

Skip connection has been widely used to bridge the high-level change-related feature maps with the corresponding bitemporal low-level feature maps, enhancing the capability for pixelwise prediction within the encoder–decoder framework. However, the information on the other scales, which are also useful to change detection, has not been fully exploited by these single-scale connections in these models based on U-Net [53]. Instead of a plain skip connection, we proposed a module of dense skip connections to integrate full-scale feature maps generated by the T1 encoder, the T2 encoder, and the decoder.

As shown in Fig. 2, we take F^3_{De} as an example to illustrate how the feature map is generated in the decoder. The low-level feature maps ($F^1_{\mathrm{En}_T1}$, $F^1_{\mathrm{En}_T2}$, $F^2_{\mathrm{En}_T1}$, and $F^2_{\mathrm{En}_T2}$) are downsampled to a scale the same as $F^3_{\mathrm{En}_T1}$ using max-pooling layers with the strides of 4 for $F^1_{\mathrm{En}_T1}$ and $F^2_{\mathrm{En}_T1}$ and 2 for $F^1_{\mathrm{En}_T2}$ and $F^2_{\mathrm{En}_T2}$, respectively. The downsampled

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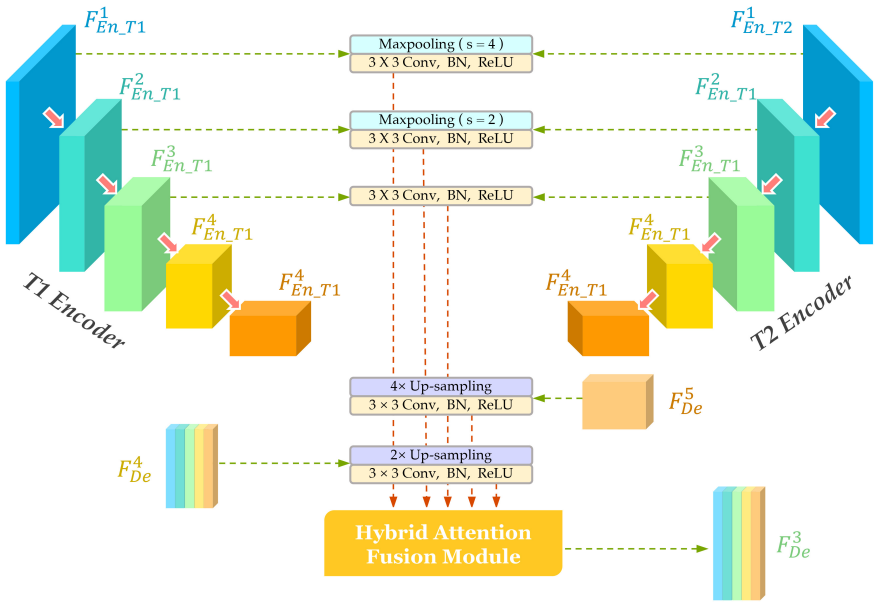

Fig. 2. Diagram that illustrates the process of dense skip connection taking the third decoder layer F^3_{De} as an example.

feature maps are then connected to the decoder. Similar to the plain skip connection in U-Net, the feature maps $F^3_{\mathrm{En_}T1}$ and $F^3_{\mathrm{En_}T2}$ in the same stage of the encoder are passed into the decoding stage directly. The high-level feature maps that contain abundant semantic information from previous decoder layers F^5_{De} and F^4_{De} are then transmitted to 4× and 2× bilinear upsampling layers, respectively, such that the generated maps could maintain the image scale. A convolution block based on the Conv-BatchNorm-ReLU architecture is appended to each combination of the pooling and upsampling layers. The processed feature maps represented by X^i_{De} can be generalized as follows:

$$F^i_{\mathrm{De}} = \begin{cases} \mathrm{HAM}\left(\mathcal{C}\left(F^i_{\mathrm{En}_{T1}}, F^i_{\mathrm{En}_{T2}}\right)\right), & i = N \\ \mathrm{HAM}\left(\begin{matrix}\mathcal{C}\left(\mathcal{D}\left(F^k_{\mathrm{En}_{T1}}, F^k_{\mathrm{En}_{T2}}\right)\right)_{k=1}^{i-1} \\ \mathcal{C}\left(F^i_{\mathrm{En}_{T1}}, F^i_{\mathrm{En}_{T2}}\right), \mathcal{C}\left(\mathcal{U}\left(F^k_{\mathrm{De}}\right)\right)_{k=i+1}^{N}\end{matrix}\right) \\ \qquad\qquad i \in \{1, 2, \ldots, N-1\} \end{cases} \tag{1}$$

where i represents the downsampling layer in the encoder, N denotes the total number of the feature scales, HAM represents the operation of the hybrid attention fusion module, $\mathcal{C}(\cdot)$ denotes a convolution operation followed by the BatchNorm and ReLU activation functions, $\mathcal{D}(\cdot)$ denotes the downsampling operation, and $\mathcal{U}(\cdot)$ denotes upsampling operation.

Dense skip connection is able to integrate detailed spatial information and abstract semantic information from multiple scales, thereby ensuring that the decoder can fully exploit multiscale features obtained at each individual stage to improve change detection. It helps narrow the semantic gap between the encoder and the decoder, and acts as an important factor to generate high-quality intermediate change maps in the refinement module.

D. Hybrid Attention Module

With multiple pairs of bitemporal features and high-level decoder features with the same size passed by dense skip connection, there is a need to fuse feature maps from different domains effectively and enhance discrimination between changed and unchanged areas. We design an HAM to aggregate these multiscale and bitemporal feature maps. There are two critical designs in HAM, including ESTAM and CAM.

The structure of HAM is illustrated in Fig. 3(a). The multiscale features $F_{\mathrm{En_}T1}$ and $F_{\mathrm{En_}T2}$ as delivered by dense skip connection are forwarded to ESTAM to capture long-range spatial–temporal dependencies. The output features are then fed into two convolutional blocks to reduce the dimension and generate enhanced features with the size the same as $F_{\mathrm{En_}T1}$ and $F_{\mathrm{En_}T1}$. The enhanced features are concatenated with the decoder features and then forwarded to CAM in order to model channelwise long-range contextual information. The residual shortcut connection integrates the original features and channel-refined features, and helps facilitate the error backpropagation and improve the model performance. The output of HAM exploits spatial, temporal, and channel context information and, thus, helps the model enhance the discriminative features for identifying changed objects or areas.

The structure of ESTAM is illustrated in Fig. 3(b). Contextual information plays an important role in remote sensing change detection tasks. The spatial–temporal attention module used a self-attention module to extract spatial–temporal context information from bitemporal features. The idea is to concatenate the maps in spatial dimension and then feed the processed maps into the self-attention module for modeling long-range spatial–temporal dependencies. The method has been proven effective in exploiting the spatial–temporal relationship among pixels, and thus, it helps produce features that are robust to the variance of illumination and

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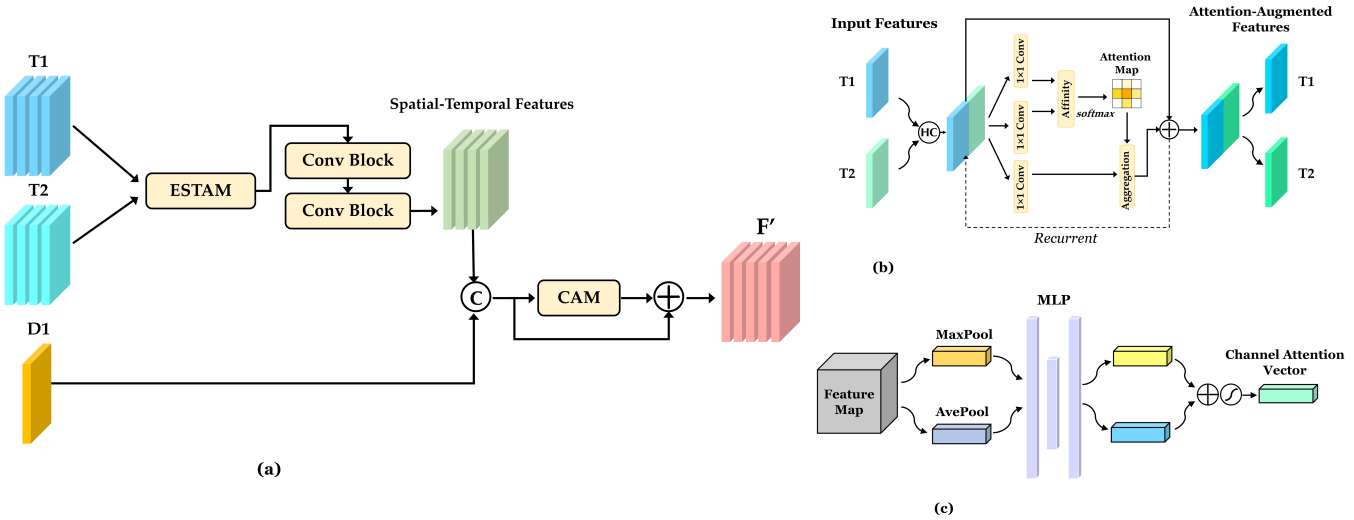

Fig. 3. Structures of attention mechanism modules are shown for (a) HAM, (b) ESTAM, and (c) CAM.

misregistration [45]. ESTAM shares the basic idea with this module. However, the self-attention mechanism used in the spatial–temporal attention module tends to generate large and complex attention maps to account for the relationships between every two pixels and, thus, requires large GPU memory to save and calculate these large attention maps. As the change detection tasks with dense nonlocal-based calculation demand computing resources, especially the graphic process unit, we introduced crisscross attention in ESTAM to deal with computation resource demands and model long-range spatial–temporal relationships in the full-scale connection module.

The bitemporal features $\{F_{T1}, F_{T2}\} \in \mathbb{R}^{C\times H\times W}$, where C represents the number of channels, and H and W represent the height and the width of feature maps, respectively, were fed into ESTAM. The pair of the features was stacked horizontally to produce a new features tensor $X \in \mathbb{R}^{C\times H\times W\times 2}$, which was passed through three convolutional layers with a 1×1 kernel size to generate three new feature maps: query matrix X_Q, key matrixX_K, and value matrix X_V, where $\{X_Q, X_K, X_V\} \in \mathbb{R}^{C\times H\times W\times 2}$. For a position p in the spatial dimension of X_Q, its feature vector is denoted as $V_p \in \mathbb{R}^C$, and we also extract the feature vectors $\psi_p \in \mathbb{R}^{(H+W\times 2-1)\times C}$ from X_K, which shares the same row and column with p. We obtain the attention vector for position p based on the matrix multiplication and the softmax operation over the channel dimension as follows:

$$A_p = \text{Softmax}\left(\psi_p \times V_p\right) \tag{2}$$

where $A_p \in \mathbb{R}^C$ represents the similarities between p and the other pixels in the same row or column.

For each pixel in X_Q, we calculate all the crisscross attention vectors and construct full-image attention maps $A_{\text{CC}} \in \mathbb{R}^{(H\times W)\times(H+W\times 2-1)}$. The attention map then multiplies the value matrix X_V to aggregate contextual information such that we obtain the aggregated map $M \in \mathbb{R}^{C\times H\times W}$ as follows:

$$M_p = A^p_{\text{CC}} \times \phi_p \tag{3}$$

where M_p denotes the aggregated features vector in a position p, A^p_C is the attention vector in p, and ϕ_p is the corresponding feature vectors in X_V, which has the same row and column as p.

The aggregated feature map is then added with X to generate the attention-augmented features $F_{\text{CC}} \in \mathbb{R}^{C\times H\times W\times 2}$, which contains contextual information for the dense prediction task. However, this basic crisscross attention module only captures contextual information in the crisscross direction but ignores information that is not situated at the crisscross path. To exploit full-image contextual information from all pixels and obtain dense contextual information, we forward the generated attention-augmented features F_{CC} to the crisscross attention module again and generate the final output features of ESTAM as $F_{\text{ESTAM}} \in \mathbb{R}^{C\times H\times W\times 2}$. With the help of the recurrent structure, global contexts can be aggregated in an indirect way, which boosts the performance of the network without extra parameters. F_{ESTAM} is then separated into two attention-enhanced feature maps $\{F^{T1}_{\text{ESTAM}}, F^{T2}_{\text{ESTAM}}\} \in \mathbb{R}^{C\times H\times W}$.

The structure of CAM is illustrated in Fig. 3(c). CAM was introduced to refine the spatial–temporal attention features and high-level decoding features. It captures long-range contextual information in the channel dimension. By multiplying input features with channel attention vector, channels relevant to land surface changes are strengthened, while irrelevant channels are weakened. First, two pooling operations (average-pooling and max-pooling operations) are employed to squeeze the spatial information in each channel, generating average-pooled features and max-pooled features. Second, both features are forwarded to a multilayer perceptron (MLP), in which the channel of features was first reduced and then restored to the size same as inputs. The channel attention vector is obtained by elementwise summation of the outputs from MLP and a sigmoid function σ. Specifically, the channel attention vector can be computed as follows:

$$V^F_{\text{CA}} = \sigma\left(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))\right). \tag{4}$$

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The extracted channel attention vector is assigned to the input feature map F using elementwise multiplication, generating the improved features F_{CA} as follows:

$$F_{\mathrm{CA}} = F \otimes V_{\mathrm{CA}}^{F}. \tag{5}$$

E. Recurrent Refinement Module

At the end of the decoder, we use a refinement module to predict the change map in a recurrent manner instead of a common method that only executes label prediction in the last layer of a network. The module makes up of several refinement units, and here, we take the third unit as an example. As discussed above, discriminative and informative features F_{De}^{3} are obtained by feeding full-scale features into HAM. The feature map is not only used in the rest stage of the decoder recursively but also used to guide the refinement of the coarse change map predicted in the former stage for producing fine change maps. Suppose that f_s denote the sth stage change map in the decoder, and f_2 is the predicted change map in the second stage. The units learn to integrate the information from the coarse label f_2 and the attention features F_{De}^{3} as inputs and then generate a finer change map. In order to produce fine change maps, f_2 is upsampled with the factor of 2 such that the spatial dimension of the generated features $f_{2\mathrm{us}}$ is the same as that of F_{De}^{3}. $f_{2\mathrm{us}}$ is then concatenated with F_{De}^{3} directly, and the concatenated feature is fed into a 3×3 convolution layer followed by a sigmoid function to produce the refined detection map f_3. Note that the refined change map is recursively fed into the next refinement unit to produce much finer results. The operations can be generalized as follows:

$$f_s = \sigma\left(\mathcal{C}\left(\mathrm{Conc}\left(F_{\mathrm{De}}^{s}, \mathcal{U}(f_{s-1}, 2)\right)\right)\right) \tag{6}$$

where σ denotes the sigmoid function, $\mathcal{C}$ denotes the convolution layer, Conc is the concatenation in channel dimension, and $\mathcal{U}(\cdot)$ is the bilinear upsampling operation with a factor of 2.

The RRM produces five change maps with increasing spatial dimensions in the decoder. To assure the progress of the coarse-to-fine adjustment, deep supervision was deployed in each refinement unit. To measure the loss between change maps and reference maps, these change maps were eventually resized to the full-resolution change map by the bilinear upsampling operation as follows:

$$F_s = \begin{cases} f_s, & s = N \\ \mathcal{U}\left(f_s, 2^{N-s}\right), & s \in \{1, 2, \ldots, N-1\} \end{cases} \tag{7}$$

where N is the number of refinement stages, $\mathcal{U}(\cdot)$ is the upsampling operation, and 2^{N-s} is the resized factors. By recursive refinement and multiscale supervision, the finer feature maps from the attention module help recover image details, and the change map gets progressively refined throughout the decoder.

F. Loss Function

For remote sensing change detection, the number of negative pixels is much greater than that of positive pixels, resulting in a serious class imbalance problem for network training. The issue could lead to insufficient optimization, which easily gets trapped in the local minima of the objective function, yielding a network severely biased toward the majority. To mitigate the negative impact of class imbalance, we apply a hybrid objective function that aggregates both binary cross-entropy loss (BCELoss) and dice loss as follows:

$$L_{\mathrm{hybrid}} = L_{\mathrm{bce}} + L_{\mathrm{dice}}. \tag{8}$$

To describe the DCELoss and dice loss, we define the predicted change map $\hat{Y}$ and the reference map Y as a set of pixels as $\hat{Y} = \{\hat{y}_i, i = 1, 2, \ldots, N\}$ and $Y = \{y_i, i = 1, 2, \ldots, N\}$. $\hat{y}_i$ denotes the probability of change in the ith pixel, and y_i denotes the reference value in the ith pixel, whereas 0 denotes the unchanged pixel and 1 denotes the changed pixel; N is the total number of pixels in a change map. BCELoss has been widely used in binary classification and segmentation tasks, and it is suitable for the binary change detection task. This objective function is formulated as follows:

$$L_{\mathrm{bce}} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log_2 \hat{y}_i + (1 - y_i)\log_2(1 - \hat{y}_i). \tag{9}$$

The dice loss is adopted to address the problem of class imbalance and improve the change detection performance. The loss first calculates a dice coefficient between $\hat{Y}$ and Y to measure the similarity of these maps. The larger the value is, the more similar they are. We need to minimize the objective function when training the network; we, therefore, subtract the dice coefficient from one for optimization as follows:

$$L_{\mathrm{dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum \hat{y}_i}. \tag{10}$$

Due to deep supervision applied in the RRM, we compute weighted hybrid loss in each of the predicted change maps as follows:

$$L_{\mathrm{total}} = \sum_{s=1}^{S} \lambda_s L_s \tag{11}$$

where S denotes the number of supervised stages, L_s denotes the hybrid loss computed in the sth stage, and λ_s denotes the weight of loss for the sth stage.

IV. Experiment Design

A. Datasets

In order to evaluate the performance and robustness of DARNet, comparative experiments were conducted on three challenging datasets, including the season-varying change detection (SVCD) dataset [54], the Sun Yat-sen University change detection (SYSU-CD) dataset [16], and the Learning Vision and Remote Sensing Laboratory building change detection (LEVIR-CD) dataset [45]. Fig. 4 presents the examples from the datasets.

SVCD is a commonly used dataset when evaluating the deep-learning-based models for remote sensing change detection. The dataset comprises seven pairs of the season-varying image obtained from Google Earth. The spatial resolution

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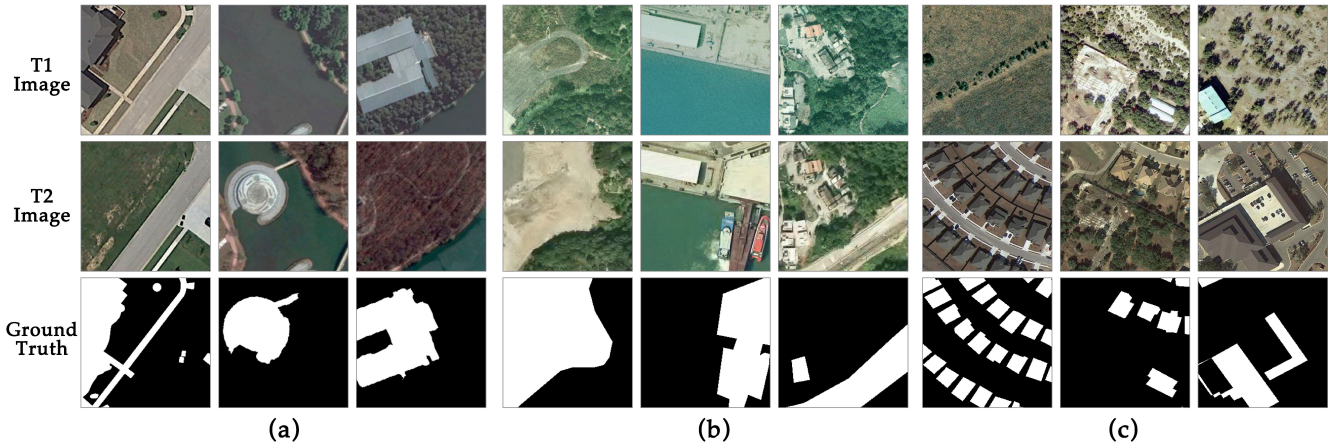

Fig. 4. Examples of bitemporal optical images and change maps from (a) SVCD, (b) SYSU-CD, and (c) LEVIR-CD.

of this dataset varies from 3 to 100 cm per pixel. In this dataset, the object changes due to seasonal variation were ignored. By cropping and random rotation, the original images were cropped to small patches with a size of 256 × 256. The generated 16 000 image pairs were further separated into training (10 000 pairs), testing (3000 pairs), and validation sets (3000 pairs).

SYSU-CD is a newly developed large-scale change detection dataset. It consists of 20 000 pairs of 256 × 256 size with the spatial resolution of 0.5 m generated by applying augmentation on the original 800 image pairs taken between the years of 2007 and 2014 in Hong Kong. These image pairs were further divided into training, testing (4000 image pairs), and validation sets (4000 image pairs). It is a challenging dataset for change detection because it contains complicated types of changes.

LEVIR-CD contains 637 image pairs with the size of 1024 × 1024 pixels and the spatial resolution of 0.5 m via the Google Earth application programming interface. The dataset mainly focuses on changes in buildings, including newly constructed buildings and destructed buildings, and 31 333 changed buildings were annotated in the label data. The authors have randomly divided the data into training (445 image pairs), testing (128 image pairs), and validation sets (64 image pairs). We further cropped the data into 256 × 256 patches without overlapping because of the limited GPU memory in our workstation. We filtered out invalid patches and obtained 2406 pairs, 731 pairs, and 333 pairs for training, testing, and validation, respectively.

B. Implementation Details

The proposed DARNet was implemented using the Pytorch framework and trained from scratch. The networks were initially set using the technique introduced in the work of He *et al.* [55]. Data augmentation, including random flipping, random rotation, and Gaussian blurring, was used to enhance the model's robustness in the training phase. Our network was trained with the Adam optimizer [56] using β_1 of 0.9, β_2 of 0.999, the weight decay of 0.01, and the batch size of 12. In terms of experiments on the SVCD, the initial learning rate was 0.002 and was then decreased by a factor of 2 every 14 epochs. We totally trained the network for 150 epochs. In terms of experiments on SYSU-CD, we set the learning rate as 0.004 and decreased it by a factor of 2 every eight epochs. We trained the model for 60 epochs in total. In terms of experiments on LEVIR-CD, we set the learning rate as 0.002 and decreased it by a factor of 2 every ten epochs. We totally trained the network for 80 epochs. It should be noted that the parameters setting used in the other tested methods are mostly consistent with the models presented in the original literature. All the experiments in the research were conducted on the workstation with Intel Xeon CPU E5-2680 v3 cores, 128-GB RAM memory, and NVIDIA Geforce RTX 2080Ti GPU.

C. Comparative Methods

The experimental results of the SOTA change detection models, including the fully convolutional early fusion (FCEF) model [37], the fully convolutional Siamese concatenation (FCSC) model [37], the fully convolutional Siamese difference (FCSD) model [37], the Siamese-based spatial–temporal attention neural network (STANet) [45], the deeply supervised image fusion network (DSIFN) [39], the deeply supervised attention metric-based network (DSAMNet) [16], and the Siamese Nested UNet (SNUNet) [41], were compared on three challenging change detection datasets in order to verify the superiority of DARNet.

The FCEF model is based on the original U-Net model. To cope with the images obtained at different times, bitemporal image pairs were concatenated before they were passed into the network, which treats inputs as an image with double channels. The fused image is then input into the encoder–decoder structure of U-Net to generate the prediction results.

The FCSC model is the FCEF variant with the Siamese network for feature extraction. Instead of fusing corresponding images acquired at different times into one composite image, images are used as the inputs into separated branches with

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identical structures and shared weights to extract deep features. In the decoder, the skip connections from two streams are concatenated to fuse the features from different time phases.

The FCSD model is generally similar to the architecture of FCSC. Different from direct concatenation utilized in FCSC, the absolute values of differences between feature pairs are concatenated in the decoding stream.

STANet was based on deep metric learning and attention mechanism. A self-attention mechanism was designed to exploit rich dependencies in spatial–temporal dimension, generating more discriminative features. STANet is a deep metric-learning-based model with a contrastive loss supervising the metric learning module.

DSIFN adopts a deep feature extraction network with the same structures and parameters as the encoder. The difference discrimination network is designed to process features from two streams. To deal with spatial heterogeneity and enhance model performance, attention mechanisms and deep supervision modules are introduced in the difference discrimination network.

DSAMNet integrates CBAM into the metric module for generating more discriminative features and employs deeply supervised modules as auxiliary to enhance the learning ability of feature extractors.

SNUNet is the combination of the Siamese network and NestedUNet for alleviation of the loss of local information in deep layers. Besides, the ensemble CAM is utilized in the network to aggregate features in multiple semantic levels.

D. Evaluation Metrics

To measure the performance of models in the experiments, we used six metrics: precision, recall, F1-score, IoU, overall accuracy (OA), and kappa coefficient. Typically, in a change detection task, a high precision value indicates a low commission error, and a high recall value indicates a low omission error. F1-score, IoU, OA, and kappa act as comprehensive metrics that are indicative of general model performance, whereas high metric values represent positive model performance. The abovementioned metrics can be formulated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$F_1 = \frac{2 \cdot \text{Precison} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{15}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{16}$$

$$\text{Kappa} = \frac{\text{OA} - P}{1 - P} \tag{17}$$

where TP denotes the number of changed pixels correctly detected; TN denotes the number of unchanged pixels classified as no-change; false positive (FP) denotes the number of unchanged pixels that are wrongly predicted as changed pixels; and false negative (FN) denotes the number of changed pixels that are wrongly classified. P in kappa denotes the hypothetical probability of chance agreement between the reference data and predictions [57], which can be formulated by

$$P = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{FN} + \text{TN})(\text{TP} + \text{TN})}{(\text{TP} + \text{FP} + \text{TN} + \text{FN})^2}. \tag{18}$$

TABLE I
QUANTITATIVE RESULTS ON THE SVCD DATASET

Models	*Pre*	*Rec*	*F1*	*IoU*	*OA*	*Kappa*
FCEF	89.87	66.49	76.43	61.85	94.74	73.75
FCSD	88.62	74.78	81.11	68.23	95.53	78.76
FCSC	89.50	73.67	80.82	67.81	95.52	78.47
STANet	87.35	95.77	91.36	84.10	97.86	90.15
DSIFN	90.19	91.33	90.76	83.07	97.80	89.51
DSAMNet	90.16	98.42	94.10	88.87	98.41	93.19
SNUNet	96.33	96.00	96.16	92.61	99.09	95.65
DARNet	97.05	96.91	96.98	94.14	99.29	96.58

V. RESULTS

A. Results for SVCD

Table I summarizes the quantitative comparisons on the SVCD dataset. As reported in the table, DARNet achieves the best performance among the tested models as it obtained the highest precision (97.05%), recall (96.91), F1-score (96.98%), IoU (94.14%), OA (99.29%), and kappa (96.58%). As for the other methods, FCEF exhibits poor performance with the worst F1-score of 76.43%, followed by FCSC with an F1-score of 80.82% and FCSD with an F1-score of 81.11%. The comparison indicates that the Siamese network could improve the overall model performance on change detection. The SNUNet achieves the second-best performance on change detection with the kappa of 95.65% due to the fact that dense connection and attention mechanism help improve the capability of feature learning. Compared with SNUNet, DARNet achieved improvements in model performance with 1.53% increases in IoU and 0.93% increases in kappa.

To qualitatively evaluate the proposed DARNet, five scenes of detection results from the testing set of SVCD are presented in Fig. 5. As shown in the figure, DARNet generates better results visually than the other models. Specifically, when detecting small changed areas, DARNet outperforms the other comparative models due to the use of the dense connection designed for multiscale detection. FCEF, FCSC, and FCSD only identify large changed areas and tend to ignore fine changes of tiny objects in the remote sensing images, such as cars and thin roads, leading to high commission errors and low recall rates in the results. With the help of an attention mechanism, DSIFN could detect the rough position of small changes, but the detected changed areas are still incomplete and inaccurate. STANet and DSAMNet that account for metric learning successfully extracted the changes at variant scales, but fine details, such as the boundaries of objects, are coarse and supersaturated compared to the ground truth.

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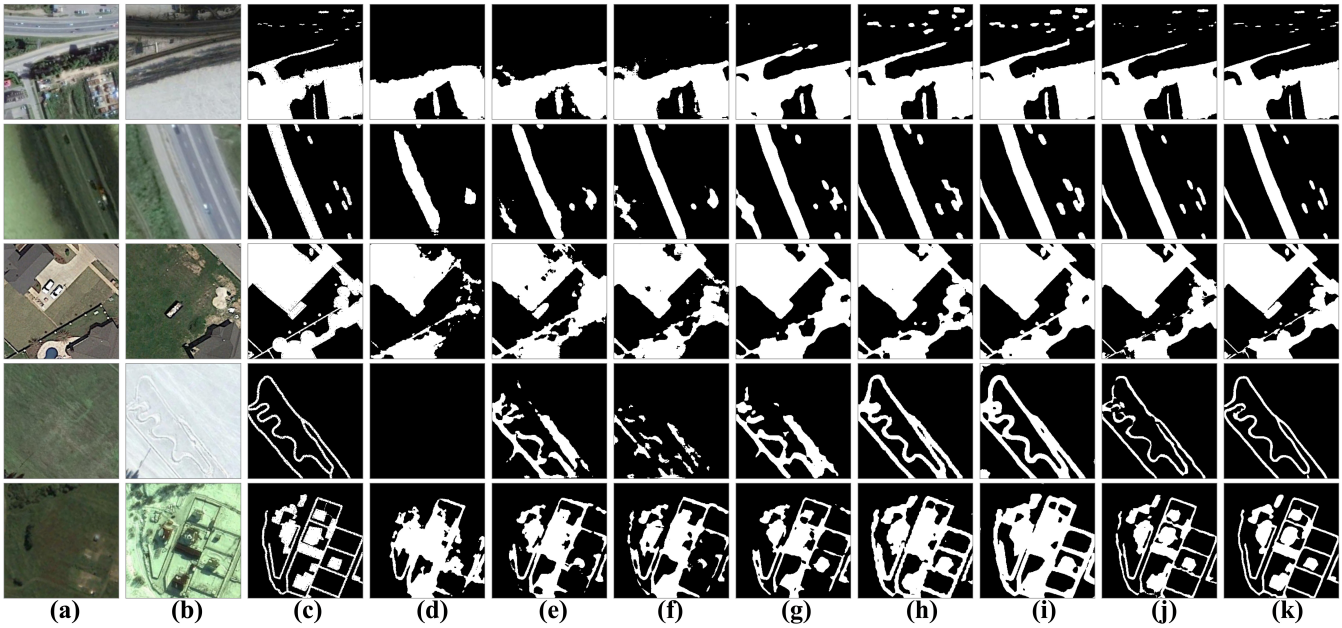

Fig. 5. Comparisons between DARNet and the other benchmark algorithms on the SVCD dataset are shown for (a) Time1 images, (b) Time2 images, (c) reference maps, and detection maps generated by (d) FCEF, (e) FCSC, (f) FCSD, (g) DSIFN, (h) STANet, (i) DSAMNet, (j) SNUNet, and (k) DARNet.

SNUNet shows comparable change detection results, but some small samples are not well detected. In addition, in terms of connectivity of change objects, SNUNet performs worse than DARNet. For large area detections, DARNet preserves fine-grained details and high compactness visually, and the generated change maps are highly consistent with the reference change maps.

With respect to the pseudochange problem, pseudochange mainly refers to the phenomenon that negative sample (nonchange area) is misclassified as a positive sample (change area) during inferencing, which tends to produce low precision and high recall in quantitative results [58]. As can be seen from Table I, STANet and DSAMNet suffer from serious pseudochange problems because recall is much higher than precision. DSIFN also generates higher recall than precision with a difference of 1.14%. Although attention mechanism has been leveraged in these networks, the extracted features are still not enough to solve the pseudochanges problem due to the insufficient exploitation of information. Among the eight networks, DARNet not only achieves the best comprehensive accuracy but also gains the best precision, which means that the proposed network is more capable of tackling pseudochange problems than other methods. As shown in Fig. 5, a pseudochange problem can be clearly observed in the neighboring area of the change objects for the STANet, DSAMNet, and DSIFN. By contrast, DARNet generates precise detection with less misclassified negative samples.

B. Results for SYSU-CD

Table II lists the quantitative results, and DARNet achieves the best F1-score (81.03%), IoU (68.10%), OA (91.26%), and kappa (75.35%). FCEF has the highest precision with 84.77%, which is slightly better than DARNet (83.04%).

TABLE II
QUANTITATIVE RESULTS ON THE SYSU-CD DATASET

Models	*Pre*	*Rec*	*F1*	*IoU*	*OA*	*Kappa*
FCEF	84.77	68.63	75.85	61.09	89.84	69.51
FCSD	79.78	74.38	76.99	62.58	89.83	70.47
FCSC	83.03	71.60	76.90	62.46	89.90	70.48
STANet	70.26	86.18	77.41	63.15	88.14	69.48
DSIFN	79.59	75.58	77.53	63.31	89.67	70.83
DSAMNet	77.11	79.90	78.48	64.59	89.67	71.69
SNUNet	78.26	74.42	76.30	61.68	89.10	69.22
DARNet	83.04	79.11	81.03	68.10	91.26	75.35

STANet obtained higher recall than other methods, while its precision is the least. Our method made a better tradeoff between precision and recall, which both are higher than most of the benchmark algorithms. SNUNet performs worse in this dataset with the lowest kappa (69.22%). Note that the overall performance on SYSU-CD is worse than SVCD, which may attribute to complicated detection scenes and confusing change objects existing in this dataset.

For intuitive comparisons, the examples of results predicted by different models on the SYSU-CD are presented in Fig. 6. DARNet achieves the best change detection results, and the generated change maps preserve the complete boundaries of changed buildings and also kept high internal compactness. FCEF, FCSC, and STANet failed to extract the newly built buildings in the urban area (Row 3). The results of FCSC and FCSD suffered from obvious distortion in the outline of the changed objects (Rows 4 and 5). STANet and DSAMNet generated many false alarm pixels in the presented scenes, which is consistent with its poorest performance in the

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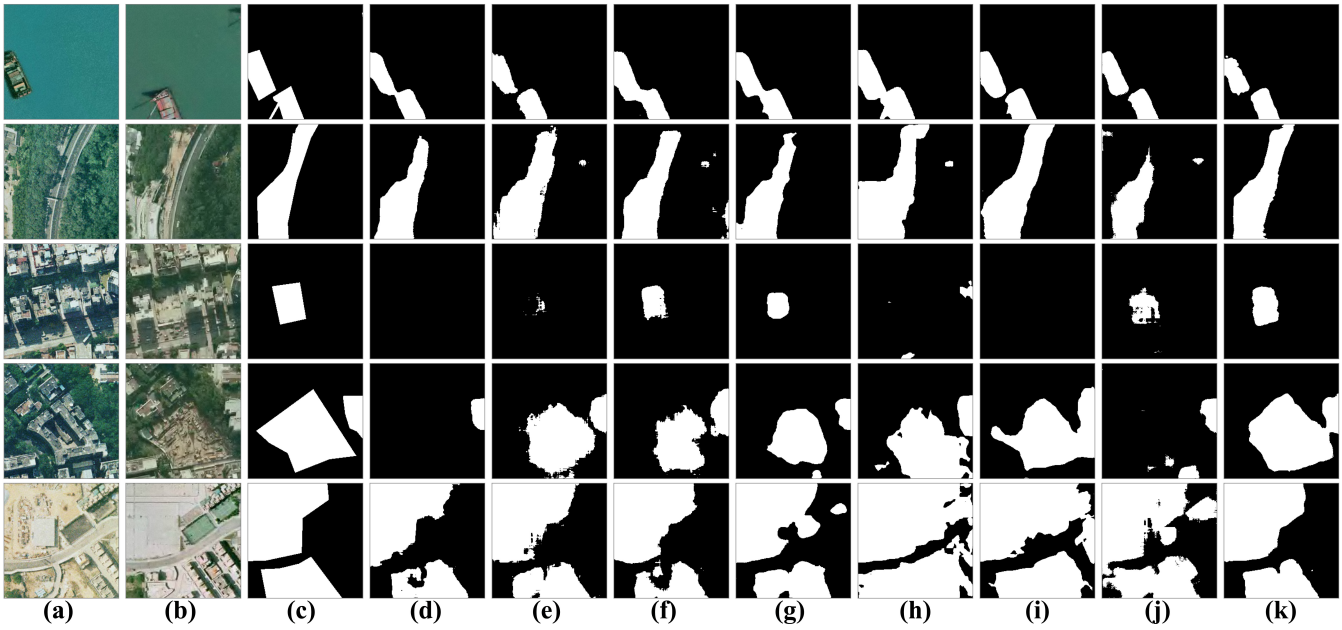

Fig. 6. Comparisons between DARNet and the other benchmark algorithms on the SYSU-CD dataset are shown for (a) Time1 images, (b) Time2 images, (c) reference maps, and detection maps generated by (d) FCEF, (e) FCSC, (f) FCSD, (g) DSIFN, (h) STANet, (i) DSAMNet, (j) SNUNet, and (k) DARNet.

statistical results. SNUNet is unable to detect some significant change areas. Both the DSIFN and DARNet could detect the changed objects and generate fewer pseudochange pixels, but DARNet is more consistent with the reference, showing more accurate extent and fewer missed detections (Rows 1 and 2).

About the performance on pseudochange, similar to Table I, the quantitative results on SYSU-CD show that the STANet and DSAMNet achieve high recall and low precision, while DARNet achieves excellent precision under the premise of best kappa coefficient. The fifth scenes of Fig. 6 demonstrate that a serious pseudochange problem appeared in the results of STANet and DSAMNet, which may attribute to the difference of illumination condition when imaging the bitemporal images. Because DARNet owns great robustness to the pseudochange problem, the result of DARNet in the same scene demonstrates that such spectral variance has limited effects on our model.

C. Results for LEVIR-CD

Compared with the results on the two datasets presented before, the LEVIR-CD only focuses on the change of building, making it easier to discriminate the change. As reported in Table III, the accuracy of tested methods is less different. The results suggested that DARNet outperformed the other benchmark methods with the best recall (91.31%), F1-score (91.98%), IoU (85.16%), OA (97.76%), and kappa (90.69%). DSIFN obtains the highest precision but low recall, indicating that it produced more missed detection in the predicted change maps. Similar to the other two datasets, FCEF achieved the worst performance among six methods due to the EF strategy. The use of the Siamese network made models address the entanglement and noncorrespondence of features in the images acquired at different times. Note that, although STANet was devised by the research group of the LEVIR-CD based on metric learning and attention mechanism, its performance is inferior to networks with the U-shaped architecture (DARNet, DSIFN, and SNUNet), which verifies the superiority of U-shaped encoder–decoder structures in the building change detection task.

TABLE III
QUANTITATIVE RESULTS ON THE LEVIR-CD DATASET

Models	*Pre*	*Rec*	*F1*	*IoU*	*OA*	*Kappa*
FCEF	89.49	84.93	87.15	77.23	96.48	85.12
FCSD	91.01	87.79	89.37	80.78	97.07	87.67
FCSC	88.30	88.60	88.45	79.29	96.75	86.56
STANet	89.48	89.89	89.68	81.30	97.09	87.99
DSIFN	93.42	89.13	91.22	83.86	97.59	89.83
DSAMNet	88.56	89.13	88.84	79.92	96.85	87.01
SNUNet	91.58	89.42	90.49	82.63	97.35	88.96
DARNet	92.67	91.31	91.98	85.16	97.76	90.69

The change maps derived by SOTA models are also shown in Fig. 7. The first three scenes illustrate the detection results of small-scale changed areas, while the other scenes show those of large-scale building changes. DARNet achieved better change detection maps compared with the tested models as it generated the change maps that are the most similar to the reference. Particularly, when detecting the crowded small-scale building groups, the prediction maps derived by DARNet have better performance than other methods. DARNet detected more individual building instances and the narrow interval between the buildings, while the buildings stick together in the results generated by other algorithms. Therefore, it significantly reduced the number of pseudochange pixels and obtained a higher precision, as computed in Table III. With regard to the large-scale building, DARNet

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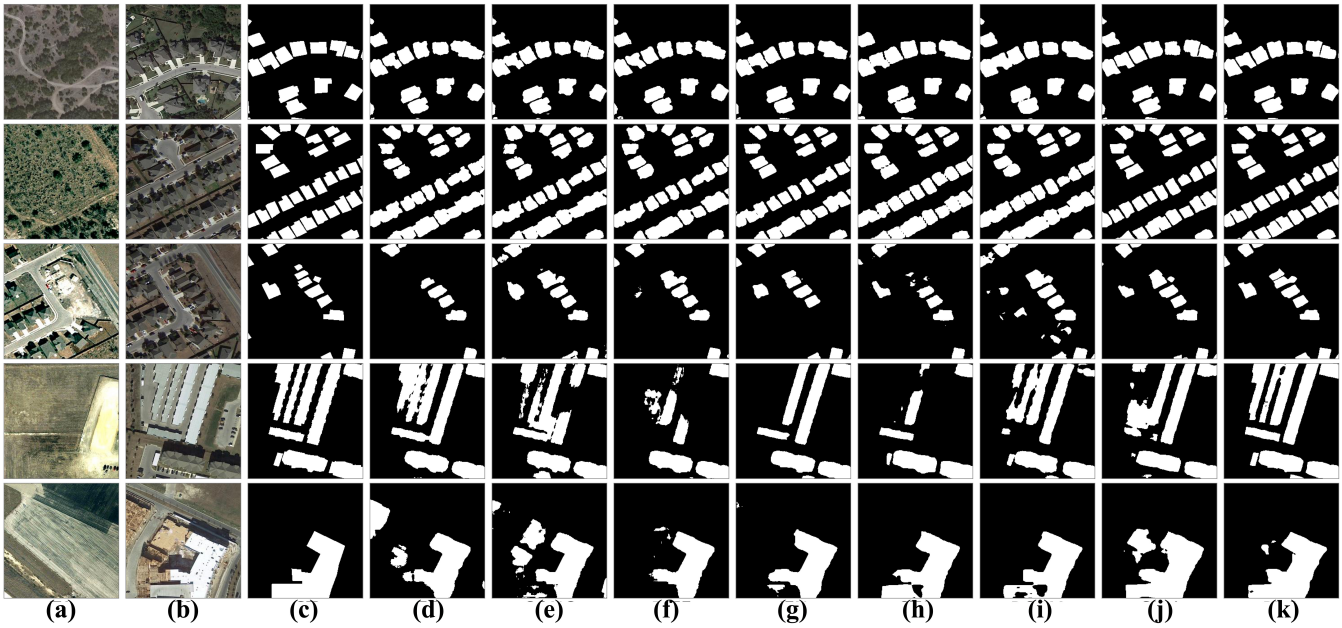

Fig. 7. Comparisons between DARNet and the other benchmark algorithms on the LEVIR-CD dataset are shown for (a) Time1 images, (b) Time2 images, (c) reference maps, and detection maps generated by (d) FCEF, (e) FCSC, (f) FCSD, (g) DSIFN, (h) STANet, (i) DSAMNet, (j) SNUNet, and (k) DARNet.

could correctly detect the complete extent of the changed area with fewer misdetection pixels and finer boundary details.

From the quantitative results, it can be seen that the pseudochange problem is not serious in LEVIR-CD datasets where each model achieves high precision and better balance with recall than SVCD and SYSUCD datasets. Thereinto, DARNet attains the second highest precision (92.67%) that is only 0.75% less than DSIFN. As shown in Fig. 7, a slight pseudochange problem caused by the variance of illumination can be observed in the third scene for DSAMNet. Moreover, another type of pseudochange problem exists in the results of the fourth scene for most comparing networks. In the fourth scene, the newly built ground shows similar characteristics with the building, and thus, these background changes are easily misclassified as foreground change, which causes the pseudochange problem. Although some pseudochange pixels are generated by our DARNet, the number of FN pixels is less than the other methods.

D. Ablation Study

To verify the validity of the key modules (DSCM, HAM, and RRM), we conducted a series of ablation experiments on three datasets. In the experiments, we mainly emphasize two indicators, IoU and kappa, which can comprehensively evaluate the performance. The baseline model is the variant of DARNet that only reserves the Siamese network, the encoder–decoder architecture, and simple skip connection. The quantitative results are listed in Table IV. As can be seen from the table, the baseline model basically yields the lowest change detection accuracies on three datasets. By plugging DSCM, HAM, or RRM into the baseline, the change detection performance of these variants is better than that of the baseline model in terms of IoU and kappa. Thereinto, the "baseline + RRM" model gains the biggest improvement among three variants, increasing the IoU and kappa by 5.86% and 4.29% on the SVCD dataset, 2.97% and 4.3% on the SYSUCD dataset, and 4.48% and 4.05% on the LEVIRCD dataset, respectively. Moreover, joint use of the proposed three modules can further improve the change detection accuracies. By combining all of the three methods, the "baseline + DSCM + HAM + RRM" model, namely, DARNet, outperforms the other models in the ablation study with the IoU improvements of 22.81%, 8.19%, and 5.76% and the kappa improvements of 15.35%, 9.00%, and 4.05% for SVCD, SYSUCD, and LEVIRCD dataset, respectively. Thereby, the substantial improvement shown in the quantitative results not only further confirms the separate effectiveness of DSCM, HAM, and RRM modules but also proves their synergy in improving change detection performance.

Notably, the most obvious improvement can be observed on the SVCD dataset due to the challenging surroundings and change types existing in the SVCD datasets We can understand the effect of the proposed modules more clearly from the results of the SVCD dataset than the other datasets. Therefore, the qualitative comparison of ablation study on SVCD datasets is shown in Fig. 8 to intuitively demonstrate the functions of these modules. As can be seen from the figure, the performance of different models generally keeps consistent with the quantitative results. To be specific, compared with the models without DSCM, the models exploiting DSCM are capable of detecting the small objects in the image, such as the cars and the bushes. Besides, the spatial details were well preserved by the integration of multiscale features in the proposed dense connection. Particularly, the derived boundaries of changed objects were more precise and complete, and the connectivity of the spline is better preserved.

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TABLE IV
QUANTITATIVE RESULTS OF ABLATION STUDY ON DARNET

Module			SVCD		SYSUCD		LEVIRCD	
DSCM	HAM	RRM	IoU	Kappa	IoU	Kappa	IoU	Kappa
			71.33	81.23	59.91	66.35	79.40	86.64
√			74.24	83.36	62.33	70.05	83.73	89.71
	√		74.02	83.16	62.48	69.23	83.56	89.56
		√	77.19	85.52	62.88	70.65	83.88	89.80
√	√		86.89	92.07	64.47	71.74	84.21	90.03
√		√	88.64	93.19	66.80	74.00	84.60	90.32
	√	√	86.43	91.76	65.57	73.10	84.07	89.92
√	√	√	94.14	96.58	68.10	75.35	85.16	90.69

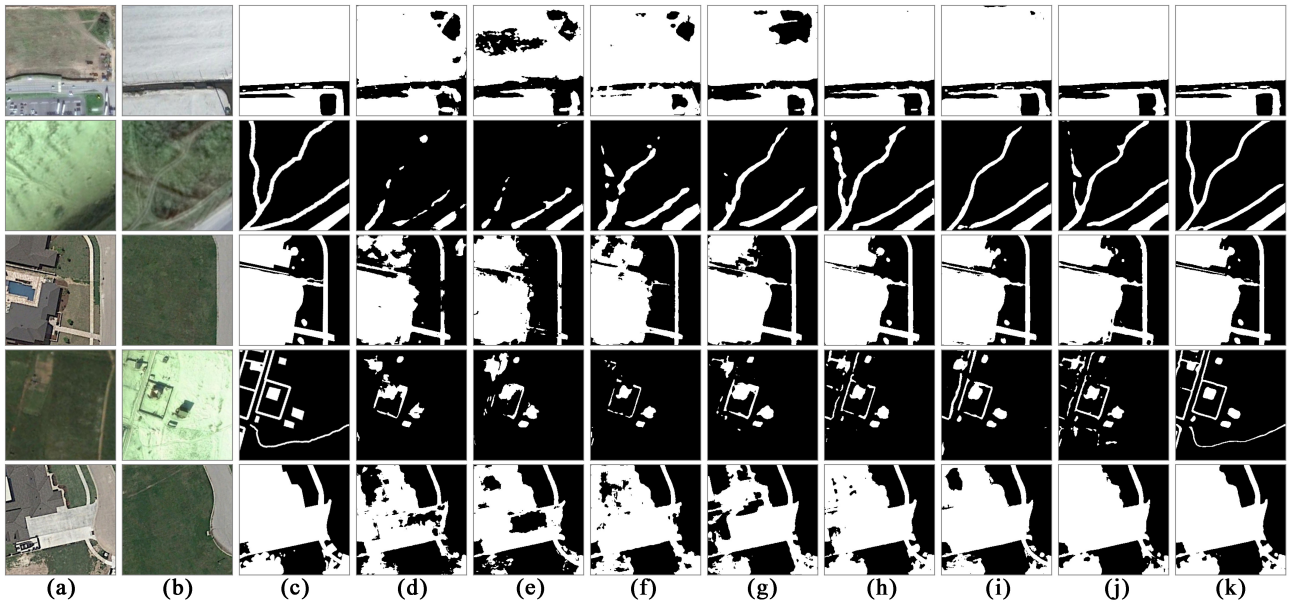

Fig. 8. Comparisons between different variants of DARNet on the SVCD dataset are shown for (a) Time1 images, (b) Time2 images, (c) reference maps, and detection maps generated by (d) baseline, (e) baseline + DSCM, (f) baseline + HAM, (g) baseline + RRM, (h) baseline + DSCM + HAM, (i) baseline + DSCM + RRM, (j) baseline + HAM + RRM, and (k) baseline + DSCM + HAM + RRM.

For the HAM module, the figure also demonstrates that the models exploiting attention mechanisms greatly decrease the omission and commission errors in the predicted change maps. Concretely, the introduction of HAM helps the model to detect more hard changed areas, which benefits from the improved discriminative ability brought by HAM. Moreover, the HAM can exploit global context information, conducive to the high internal compactness in the detected area. For a better understanding of stagewise refinement brought by RRM, we visualized the intermediate change maps of four representative examples in Fig. 9. One can observe that DARNet obtained coarse change maps in the first decoding stage, and then, the changed area gets progressively refined throughout the decoder. The results in stage 3 have been almost identical to the reference data. The change maps in stages 4 and 5 contain more accurate spatial details than the results in stage 3, such as the narrow path and the interval between objects. The quantitative comparison of the change maps derived in different stages is also computed and shown in Table V. Consistent with the visual comparison, the statistical results demonstrate that six evaluation metrics are gradually improved, and the result of the final stage reaches the optimal performance among the change maps from five stages.

TABLE V
QUANTITATIVE RESULTS OF CHANGE MAPS DETECTED IN DIFFERENT STAGES OF REFINEMENTS ON THE SVCD DATASET

Change map	Pre	Rec	F1	IoU	OA	Kappa
RRM-Stage 1	93.94	91.42	92.66	86.32	98.29	91.69
RRM-Stage2	95.78	94.66	95.22	90.87	98.88	94.58
RRM-Stage3	96.68	96.26	96.47	93.18	99.17	96.00
RRM-Stage4	96.97	96.81	96.89	93.97	99.27	96.48
RRM-Stage5	97.05	96.91	96.98	94.14	99.29	96.58

E. Complexity Analysis

To confirm the superiority of novel models, we need to evaluate their performance comprehensively. In addition

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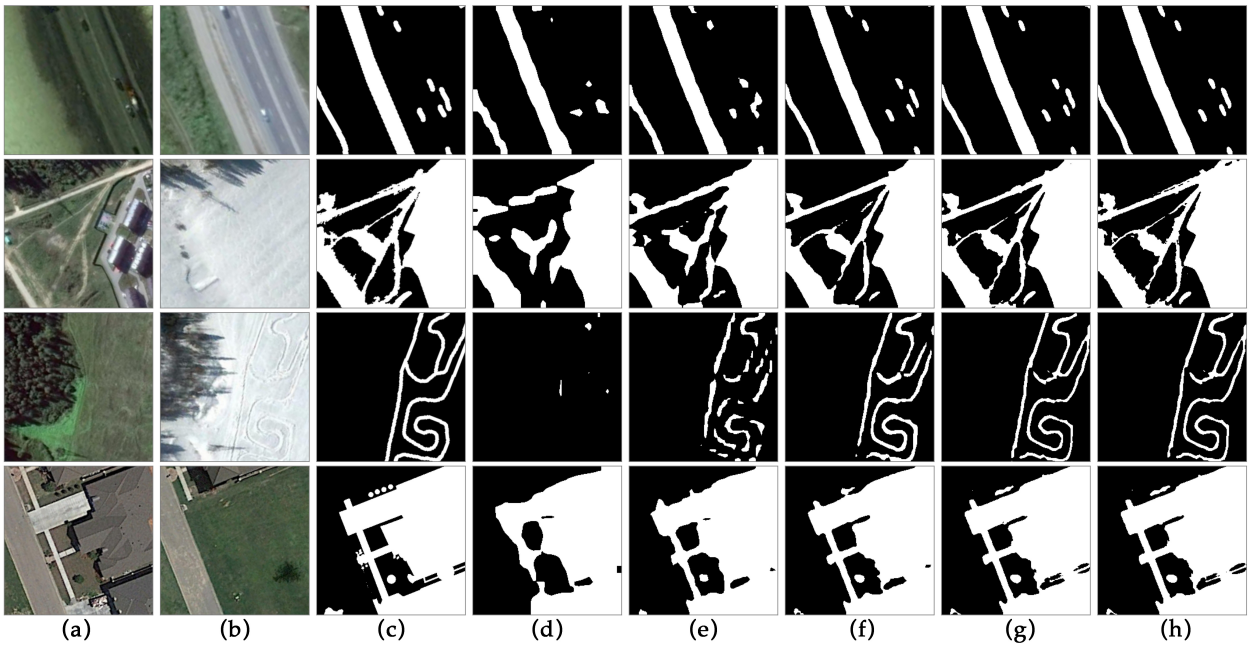

Fig. 9. Qualitative comparison of predicted results from different stages of RRM on the SVCD dataset are shown for (a) Time1 images, (b) Time2 images, (c) reference maps, and change maps of (d) Stage1, (e) Stage2, (f) Stage3, (g) Stage4, and (h) Stage5.

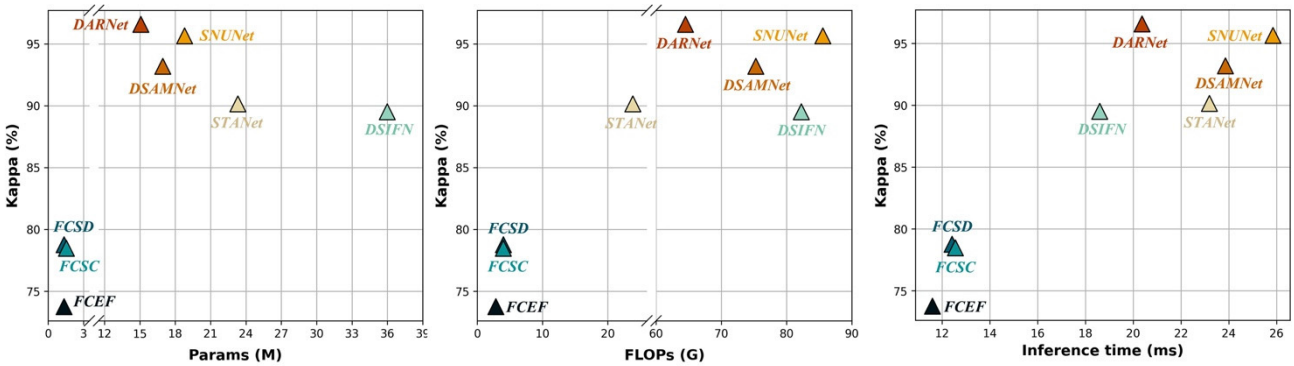

Fig. 10. Comparison of model complexity in terms of the number of parameters, the consumption of inference time, FLOPs, and kappa coefficient.

to the accuracy, the complexity of the models should also be compared. Therefore, we analyzed DARNet and other comparison networks from four aspects, including the number of parameters (Params), the consumption of inference time, floating-point operations (FLOPs), and the kappa coefficient. Fig. 10 distinctly presents the comprehensive performance of different networks. It can be seen that the proposed DARNet obtains the highest kappa and relatively low Params, FLOPs, and the cost of inference time among five improved networks. The networks with simple architecture (FCEF, FCSC, and FCSD) have lower computational cost and model complexity. However, these FC-series networks show poor accuracy because they have a weak ability of feature extraction and tend to underfit during training. Our proposed network achieves the greatest improvement by sacrificing an acceptable efficiency. Although the accuracy of SNUNet is similar to DARNet, its model complexity and computational complexity are all higher than DARNets due to the complex architecture. The specific values of the four metrics are listed in Table VI. In sum, this comparison proves that the DARNet has a better tradeoff between complexity and accuracy. Therefore, since the proposed network is a lightweight attention-based model, it has the potential to be adapted to mobile devices, such as drones and smartphones, in order to carry out fast and accurate change detection mapping.

VI. Discussion

A. Parameter Analysis on Loss Weighting

To address the problem of gradient vanishing and accelerate the convergence of the network, deep supervision is used in our network. For all the models presented before, the weights of loss in different stages keep the same (0.2 for every stage) because the deep supervision is cooperated with the stagewise refinement by RRM. The RRM uses coarse results predicted in the former layer as input features forwarded to the latter layer to conduct prediction refinement. Hence, the predictions in each stage are equally important, and thus, we set them

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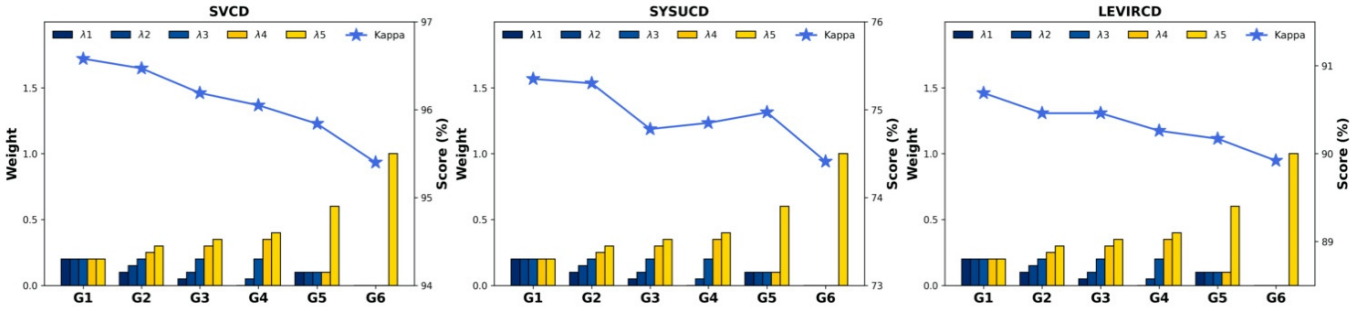

Fig. 11. Influence of loss weighting on the change detection performance in terms of kappa coefficient for three datasets.

TABLE VI
COMPLEXITY COMPARISONS OF DARNET AND OTHER BENCHMARK METHODS IN TERMS OF THE PARAMETERS, INFERENCE TIME, FLOPS, AND KAPPA COEFFICIENT

Model	Params(M)	Inference Time (ms)	FLOPs(G)	Kappa(%)
FCEF	1.35	11.60	2.85	73.75
FCSD	1.35	12.42	4.01	78.76
FCSC	1.55	12.56	4.00	78.47
STANet	23.32	23.19	23.84	90.15
DSIFN	35.99	18.60	82.26	89.51
DSAMNet	16.95	23.86	75.29	93.19
SNUNet	18.80	25.85	85.56	95.65
DARNet	15.09	20.37	64.51	96.58

as the same weight when training the models. To verify the assumption, we carry out a comparison experiment on three datasets by training the models using six different groups of weights, and the results are summarized in Table VII. It can be seen that the model trained with equal weights achieves the best performance in all datasets. Besides, the influence of loss weighting on change detection performance is not significant. The range of calculated kappa values is narrow, which only obtains 1.18% (SVCD), 2.29% (SYSUCD), and 0.77% (LEVIRCD). Moreover, Fig. 11 intuitively demonstrates that, as the variance of weights increases, the kappa coefficient experiences general downward trends in three datasets. Notably, in previous studies [16], [59], [60], scholars tend to allocate a larger weight to the output supervision in order to highlight the importance of the output layer. However, the results presented here indicated that such allocation is not always optimal. The setting of loss weights should be in accordance with the design of the network in order to fully exploit the performance of the model and obtain the best prediction.

B. Robustness to Degradation

It's common that remote sensing data suffer from various kinds of degradations, such as spectral variability, signal noise, and geometric distortion, which severely affects the feature embedding and prediction for the deep learning-based models [61], [62]. Therefore, the robustness to degradations is crucial when developing an effective algorithm for remote sensing applications. In order to evaluate the robustness of the proposed DARNet, we designed an extensive simulation experiment, which adds different types of degradations to the remote sensing images and tests the performance of trained models on these simulation datasets. As shown in Fig. 12, six types of degradation were added to the SVCD dataset, respectively. The quantitative results comparing DARNet and SNUNet are summarized in Table VIII. It can be seen that all the six kinds of degradations caused the decline of kappa. For DARNet, the "Erase" type has the largest effect (14.01%) on kappa, and the "Brightness & Contrast" type is the weakest (2.07%) among six degradations. Notably, the decline of our network affected by six degradations is far smaller than SNUNet in all of the simulation datasets. The results here clearly indicate that the proposed DARNet has better robustness to various degradation, which may attribute to discriminative and robust features produced by modeling long-range relations in three feature dimensions through HAM.

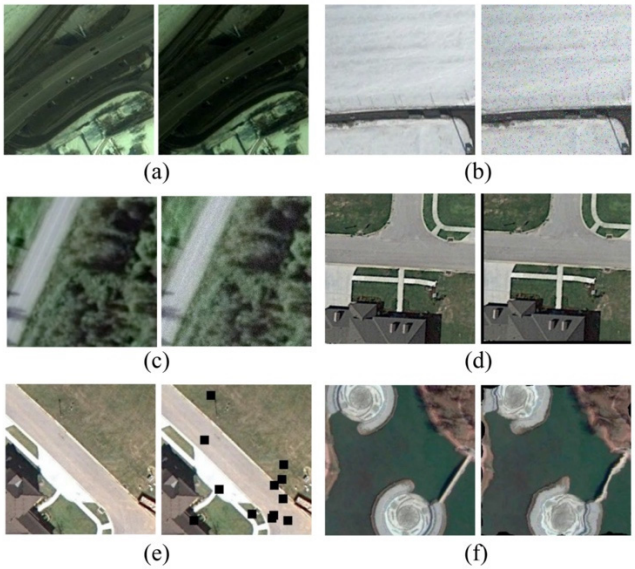

Fig. 12. Examples of six types of degradation on the SVCD dataset are shown for (a) random brightness and contrast, (b) salt and pepper noise, (c) Gaussian noise, (d) random movement, (e) random erase, and (f) elastic transform. In each group, (Left) is the original image, while (Right) is the degraded image.

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TABLE VII
QUANTITATIVE RESULTS OF PARAMETER ANALYSIS FOR LOSS WEIGHTING

Weights					SVCD		SYSUCD		LEVIRCD	
λ_1	λ_2	λ_3	λ_4	λ_5	IoU	Kappa	IoU	Kappa	IoU	Kappa
0.2	0.2	0.2	0.2	0.2	**94.14**	**96.58**	**69.59**	**76.70**	**85.16**	**90.69**
0.1	0.15	0.2	0.25	0.3	93.96	96.47	68.04	75.30	84.82	90.46
0.05	0.1	0.2	0.3	0.35	93.50	96.19	67.21	74.78	84.83	90.46
0	0.05	0.2	0.35	0.4	93.26	96.05	67.25	74.85	84.55	90.26
0.1	0.1	0.1	0.1	0.6	92.92	95.84	67.51	74.97	84.42	90.17
0	0	0	0	1	92.17	95.40	66.80	74.41	84.03	89.92

TABLE VIII
QUANTITATIVE RESULTS OF SIMULATED DEGRADATION EXPERIMENT ON THE SVCD DATASET

Degradation Type	DARNet		SNUNet	
	Kappa	Decrement	Kappa	Decrement
Raw	96.58	—	95.65	—
Brightness&Contrast	94.51	2.07	70.21	25.44
Salt&Pepper Noise	92.08	4.50	70.80	24.85
Gaussian Noise	89.10	7.48	66.56	29.09
Movement	89.74	6.84	82.89	12.76
Erase	82.57	14.01	76.39	19.26
Elastic Transform	92.13	4.45	87.08	8.57

VII. CONCLUSION

In this article, a novel DARNet was proposed for remote sensing change detection. DARNet integrates three components of the DSCM, the HAM, and the RRM. DSCM effectively aggregates full-scale features, HAM exploits abundant contextual information and enhances the representation ability for identifying the change objects, and RRM with deep supervision is designed for the stagewise refinement of the predicted change maps. The experiments were conducted on challenging datasets, and the results indicate that DARNet achieves outstanding performance for change detection in terms of accuracy and complexity and surpasses the tested SOTA models. Ablation studies further illustrate the effectiveness of the designed components. The promising performance of DARNet shows great potential for its application in detecting changes in complex scenes using very-high-resolution bitemporal remote sensing images. The source code, trained weights for DARNet, and the results of three benchmark datasets will be publicly available at https://github.com/jimmyli08/DARNet-CD for further study.

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